

Curriculum Vitae

Dr. PUSHPENDRA KUMAR

Associate Professor
Dept. Of Chemistry,
J S University
Shikohabad, Uttar Pradesh - India
E-Mail:- pushpendra.kmr1@gmail.com
Mob: 8894900779



Specialization: Physical Chemistry/ Material Science / Photocatalytic Hydrogen Generation

Career Objective: Seeking a challenging research environment where I can implement my skills to explore the different multi and inter disciplinary scopes and areas in the subjects of Chemistry and Material Science. This would be enthusiastically helpful for me to serve the society, along with continuous personal and professional development and to become a valuable asset for the organization.

Strength

My strengths include self-confidence, task oriented, time punctuality, eagerness to learn new things, hard-work and adaptability to work in various conditions as a team.

Education

Examination Passed	Year	Board/ University	% of marks
Sec	2001	U. P. Board	70.6
Sr. Sec	2003	C. B. S. E. Board	65.0
B.Sc Hons(Chemistry)	2006	Dayalbagh Educational Institute, Agra,U.P	63.2
M.Sc	2008	Dayalbagh Educational Institute,Agra,U.P	73.0
M.Phil	2009	Dayalbagh Educational Institute,Agra,U.P	79.9
Ph.D	2012	Dayalbagh Educational Institute,Agra,U.P	2012

- Solar – Hydrogen production by photoelectrochemical and photocatalytic route
- Application of nanostructured metal oxides in Spintronics, Dye degradation
- Synthesis of nanoarchitected metal oxide semiconductor
- Modified nanostructured system (Bilayered system, Quantum dot modification, Surface Modifications)
- Development of Gas sensors

Fellowship Awarded

1. UGC – SAP Research Fellowship for Meritorious Students from May 2009 to March 2012.
Awarded CSIR – Senior Research Fellowship, 2012.

2. Awarded International Travel Support by DST, New Delhi to attend the International Conference held in Houston, USA from 24 – 29 June, 2012.

Projects/ Thesis Submitted

1. M.Sc Project entitled “A Study on the Sonochemical Preparation and Characterization of Nanostructured Copper Oxide & its Application in PEC Splitting of Water”.
2. M.Phil Thesis entitled “An Emzyme Catalyzed Biohydrogen Production: A Study on Some Genetic Manipulations to Optimize Hydrogenase Expression”.
3. Ph.D Thesis entitled “An Study on Electrochemical Redox Photoprocesses for Splitting of Water Using Nanostructured Metal Oxides”.

Visits Abroad

1. Energy Research Centre, University of Maryland, College Park, USA.
2. Center for Renewable Energy Technology, Northeastern University, Boston, USA

Instrumental Experience

1. X-Ray Diffractometer (Bruker D8 Advance)
2. FE-SEM (Jeol)
3. Particle Size and Zeta Potential Analyzer (Malvern)
4. UV-Vis Spectrophotometer (Shimadzu)
5. Gas chromatography (Shimadzu)
6. Mass Spectrometer ((Shimadzu)
7. HPLC (Shimadzu)
8. Electrochemical Work Station (Autolab)
9. FTIR (PerkinElmer)
10. TGA (PerkinElmer)
11. DSC (PerkinElmer)
12. Atomic Force Microscope (Nanosurf)

Teaching Experience

Organization	Duration	Designation
Nikhil Eng. College, Agra	01/10/2012 – 29/6/2013	Assistant Prof.
Arni University, Himachal Pradesh	1/7/2013 - 3/7/2015	Assistant Prof.
Lovely Professional University, Punjab	25/7/2015 -31/12/2021	Assistant Prof.
J S University, Uttar Pradesh	5/01/2022 – Till date	Associate Prof.

Administrative Experience

1. Worked as a “Head, Dept. of Chemistry” in Arni University, Himachal Pradesh.
2. Appointed as Chairperson in Approval of “Board of Study of Dept. of Chemistry” involving the approval of syllabus of MSc (Chemsitry), BSc Hons (Chemistry) in Arni University.

3. Worked as Lab Incharge, Central Instrumentation Facility, Division of Research and Development, Lovely Professional University.

4. Worked as NAAC coordinator in Division of Research and Development, LPU

Project/Dissertation under Supervision

MPhil Thesis - 1

PhD Thesis - 1

MSc Dissertations-11

Subjects Taught

- Quantum Chemistry
- Statistical Thermodynamics
- Kinetics
- Classical Thermodynamics
- Electrochemistry
- Irreversible Thermodynamics
- Engineering Chemistry
- Analytical Chemistry
- Spectroscopy

Research Publications

Publications: 46

Citations: 812

h-Index: 13

i10-Index: 14

Research Articles

1. Bilayered nano-hetero-structured n/n junction thin-film electrodes, $\text{WO}_3/\text{Yb-Mo-BiVO}_4$, for efficient photoelectrochemical water splitting, *J Appl Electrochem*, 52 (2022) 535–558. <https://doi.org/10.1007/s10800-021-01649-8>

2. IV-curve and structural studies of the composite mixture of reduced graphene oxide and silver nanopowders, *Mater Sci: Mater Electron*, 33 (2022), 1228–1244. <https://doi.org/10.1007/s10854-021-07416-x>

3. The Comparative Study of Antibacterial Activity of Zinc Oxide and Silver Doped Zinc Oxide, *Advanced Engineering Science*, 54 (2022), 242-247

4. A Study on Effect of Thermal Treatment on Crystalline Properties of Microwave Assisted Synthesized TiO_2 , *Advanced Engineering Science*, 54 (2022), 61-70.

5. Magnetically recyclable silica-coated ferrite magnetite-K₁₀ montmorillonite nanocatalyst and its applications in O, N, and S-acylation reaction under solvent-free conditions, *RSC Adv.*, 11 (2021) 21291-21300
6. Vertically aligned ZnO nanorods for photoelectrochemical water splitting application, *Materials Letters*, 277 (2020), ID. 128295, DOI: <https://doi.org/10.1016/j.matlet.2020.128295>
7. Novel synthesis of advanced Cu capped Cu₂O nanoparticles and their photo-catalytic activity for mineralization of aqueous dye molecules, *Materials Letters*, 276 (2020) ID. 128294, DOI: <https://doi.org/10.1016/j.matlet.2020.128294>
8. Effect of Addition of Polyaniline (PANI) on the properties of copper (II) Oxide nanoparticles, *Journal of Physics: Conference Series*, 1531 (2020) 012036.
9. Structural and Optical Properties of Sr₂Co_{1.9}Bi_{0.1}Fe_{11.9}Zn_{0.1}O₂₂/PANI Nanocomposites, *Journal of Physics: Conference Series*, 1531 (2020) 012006.
10. Effect of titanium dioxide on the structural and electrical conductivity of cadmium oxide, *Materialstoday:Proceedings*, 26 (2020), 1330-1335.
11. Gas Sensing Properties of ZnO-SnO₂ Nanocomposite, *Asian Journal of Chemistry*, 32 (2020), 64-68
12. Carbon Monoxide Gas Sensor Based on Fe-ZnO Thin Film, *Asian Journal of Chemistry*, 30 (2018), 2737-2742.
13. Water mediated synthesis of pyrano [2, 3-c] pyrazoles using L-histidine as an effective catalyst, *AIP Conference Proceedings*, 1860 (2017), 020060.
14. ZnO-Fe₂O₃ heterojunction for photocatalytic degradation of victoria blue dye, *AIP Conference Proceedings*, 1860 (2017), 020065.
15. Influence of Co-doping on the Structural, Optical and Magnetic Properties of ZnO Nanoparticles, *Int. J. Nanoelectronics and Materials* 9 (2016) 1-8.
16. Photoelectrochemical Splitting of Water to Produce a Power Appetizer Hydrogen: A Green System for Future (A Short Review), *Oriental Journal of Chemistry*, 32 (2016) 1473-1483
17. Modified structural, morphological and photoelectrochemical properties of 120 MeV Ag⁹⁺ ion irradiated BaTiO₃ thin films, *Current Applied Physics*, 13 (2013), 344-350.
18. Nanostructured bilayered thin films in photoelectrochemical water splitting - A review, *International Journal of Hydrogen Energy*, 37 (2012) 18713 – 18730.
19. Photoelectrochemical water splitting with nanocrystalline Zn_{1-x}Ru_xO thin films, *International Journal of Hydrogen Energy*, 37 (2012) 12138 – 12149.
20. Synthesis and characterization of nanocrystalline Zn_{1-x}M_xO (M= Ni, Cr) thin films for efficient photoelectrochemical splitting of water under UV irradiation, *International Journal of Hydrogen Energy* 36 (2011) 4280 - 4290.
21. Vertically aligned nanocrystalline Cu – ZnO thin films for photoelectrochemical splitting of water, *J. Mater. Sci.*, 46 (2011) 3792-3801.

22. Nanostructured ZnO for photoelectrochemical splitting of water to produce hydrogen: swift heavy ion irradiation vis-à-vis dye sensitization, *Int.J. Nanoparticles*, 4 (2011) 248-267.

23. Irradiation induced modifications and PEC response - A case study of SrTiO₃ thin films irradiated by 120 MeV Ag⁹⁺ ions, *International Journal of Hydrogen Energy*, 36 (2011) 5236-5245.

24. Enhanced Photoelectrochemical Response of BaTiO₃ with Fe-Doping: Experiments and First-Principles Analysis, *Journal of Physical Chemistry C*, 115 (2011) 24373 – 24380.

25. Structure and magnetic properties of Zn_{1-x}Fe_xO nanoparticles synthesized by Sol-Gel Co-precipitation method, *IJCET*, 5 (2014) 70-73.

26. Caking and Characterizing Graphene Oxide Thin Films via Electrodeposition Technique for Possible Application in Photoelectrochemical Splitting of Water, *Proceeding of International conference on recent Trends in Applied Physics and Material Science*, AIP Conf. Proc, 1536 (2013), 537-538,.

Books

27. Laboratory Manual for Biochemical Analysis, 2022, Reyansh Authortopic Pvt Ltd, ISBN: 9789393301239

28. Laboratory Manual for Engineering Chemistry, 2022, Reyansh Authortopic Pvt Ltd, ISBN: 9789393301000

Chapters

29. Anticancer Activity of Curry Leaf (*Murraya Koenigii*), Pushpendra Kumar, Amit K. Chaturvedi, *International Conference on Global Research Trends in Higher Education*, ISBN: 978-93-91535-95-7. Edited by Dr. Manoj K. Chaudhary, 2023.

30. Antimicrobial and antioxidant activity of tomato leaves, Pushpendra Kumar, Amit K. Chaturvedi, *International Conference on Basic Sciences: Global Challenges and Solutions for Sustainable Development*, ISBN: 978-93-91535-91-9, Edited by Dr. Rajiv G. Weginwar, 2023.

31. Bacterial pigments, Amit K. Chaturvedi, Pushpendra Kumar, *International Conference on Basic Sciences: Global Challenges and Solutions for Sustainable Development*, ISBN: 978-93-91535-91-9, Edited by Dr. Rajiv G. Weginwar, 2023.

32. Biocatalyst for Industrial Chemical Transformations, Amit K. Chaturvedi, Pushpendra Kumar, *International Conference on Modern Engineering, Science & Management*, ISBN: 978-93-91535-51-3, Edited by Dr. Navanath Narawade, 2023.

33. Decontamination of Pharmaceutical Aquatic Waste Using ZnO Based Solar Catalyst, Pushpendra Kumar, Amit K. Chaturvedi, *International Multidisciplinary Conference on Recent Innovations in Science, Engineering, Management and Humanities*, ISBN: 978-93-91535-72-8, Edited by Dr. Gaurav Yadav, 2023.

34. Food Preservatives- A Perspective, Amit K. Chaturvedi, Pushpendra Kumar, *International Conference on Modern Engineering, Science & Management*, ISBN: 978-93-91535-51-3, Edited by Dr. Navanath Narawade, 2023

35. Productions of secondary metabolites and their functions in plants, Pushpendra Kumar, Amit K. Chaturvedi, International conference on recent developments in Engineering, Science, Humanities and Management, ISBN: 978-93-91535-43-8, Edited by Dr. Ravi Gupta, 2023
36. Pyrazole Compounds Synthesis Strategies and Their Pharmacological Applications, Sumeet Kumar, Pushpendra Kumar, 3rd International Multidisciplinary Conference on Recent Innovations in Science, Engineering, Management and Humanities, ISBN: 978-93-91535-99-5, Edited by Dr. Gaurav Yadav, 2024.
37. Renewable Energy for Agriculture, Amit K. Chaturvedi, Pushpendra Kumar, International conference on Science, Engineering & Technology Innovations, ISBN: 978-93-91535-83-4, Edited by Dr. Nilesh J. Uke, 2023.
38. Role of Mobile in Agriculture Sector, Amit K. Chaturvedi, Pushpendra Kumar, International Conference on Basic Sciences: Global Challenges and Solutions for Sustainable Development, ISBN: 978-93-91535-91-9, Edited by Dr. Rajiv G Weginwar, 2023.
39. Solid Waste and Its Management, Amit K. Chaturvedi, Pushpendra Kumar, International Conference on Modern Engineering, Science & Management, ISBN: 978-93-91535-51-3, Edited by Dr. Navanath Narawade, 2023
40. Structure and Characterization of Graphene Oxide, , Amit K. Chaturvedi, Pushpendra Kumar, International conference on Science, Engineering & Technology Innovations, ISBN: 978-93-91535-83-4, Edited by Dr. Nilesh J. Uke, 2023.
41. Synthesis and Pharmacological Evaluation of Pyrazole Derivatives: A Comprehensive Review, 3rd International Multidisciplinary Conference on Recent Innovations in Science, Engineering, Management and Humanities, ISBN: 978-93-91535-99-5, Edited by Dr. Gaurav Yadav, 2024.
42. Various Types of Acne and Their Therapeutic Treatment, International conference on Multidisciplinary Innovative Research and Studies, ISBN: 978-93-91535-45-2, Edited by Dr. Gaurav Yadav, 2023.
43. Hydrogen Generation Via Photoelectrochemical Splitting of Water. (2018) In In: Martínez, L., Kharissova, O., Kharisov, B. (eds) Handbook of Ecomaterials. Springer, Cham. https://doi.org/10.1007/978-3-319-48281-1_33-1
44. Synthesis and Characterization of Fe₂O₃-ZnO Nanocomposites for Efficient Photoelectrochemical Splitting of Water. Chapter 42, Section B- Energy Perspectives, pp 229-232, Chemistry of Phytopotentials: Health, Energy and Environmental Perspectives. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-23394-4_48
45. A Clean and Green Hydrogen Energy production using Nanostructured ZnO and Fe - ZnO via Photoelectrochemical Splitting of Water. Chapter 41, Section B- Energy Perspectives, 191-195, Chemistry of Phytopotentials: Health, Energy and Environmental Perspectives. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-23394-4_40
46. Synthesis and Characterization of Nanostructured ZnO for Solar Hydrogen Generation: Effect of Electrolyte pH on Photoelectrochemical Application, Proceeding, International Conference on Nanoscience and Technology in Chemistry, Health, Environment and Energy 2010 (NATCHEE 2010), Mc Graw Companies, ISBN: 0-07-068063-9, Tata McGraw Hill Education Private Limited, 2010

Attended Conferences/ Workshops

1. Participated in workshop on “Paradigm Shift in Research and Development”, 15th September 2023, organized by ICAFI Business School
2. Presented a paper entitled “Photoelectrochemical Splitting of Water into Hydrogen: The Sustainable Energy Carrier for Future” 106th Indian Science Congress (ISC 2019), Lovely Professional University, Punjab.
3. Participated in British Council Researcher Links Workshop on “Generation, Storage and Utilisation of Renewable Hydrogen: Towards Sustainable Energy”, 30th to 5th December 2018 organized by British Council, Agra.
4. Participated in National Workshop on Environmental Sciences, 10th January 2018 at Lovely Professional University, Punjab
5. Participated in NPTEL Workshop organized by IIT Madras on 12th October 2017 at Lovely Professional University, Punjab
6. Presented a paper entitled “Photoelectrochemical Splitting of Water using Dye-Sensitized ZnO Thin Films” in National Conference on Current Advances in Science and Technology (CAST-2016), Rama University, UP.
7. Presented a paper entitled “ZnO-Fe₂O₃ heterojunction for photocatalytic degradation of Methylene blue dye”, International Conference on Recent Advances in Fundamental and Applied Sciences (RAFAS 2016), Lovely Professional University, Punjab.
8. Presented a paper entitled “Nano Approach towards Power Apetizer” in workshop on Recent Trends in Engineering & Technology, 26 February, 2015.
9. Presented a paper entitled “A Study on Photoelectrochemical behavior of Dye Sensitized Photoelectrochemical cell based on transparent conductive ZnO”, International Multi track Conference on Sciences, Engineering & Technical innovations, IMTC (2014), CT Group of Institution, Jalandhar, Punjab.
10. Presented a paper entitled “Photoelectrochemically Spiltting of Water into Hydrogen: The Sustainable Energy Carrier of the Future”, National Conference on Recent Advances in Chemical and Environmental Sciences- RTCES 2014, Arni University, Himachal Pradesh
11. Participated in National Symposium on “Advanced Materials” held on 27-28 May 2014 at IIT Mandi, HP, India.
12. Presented a paper entitled as “Hydrogen- A Green Energy Carrier” in National Conference on “Advances in Basic and Applied Sciences” held on 10 May 2014 at Career Point University, HP, India
13. Participated in Workshop entitled as “National Workshop on Advances in Biotechnology”, 24 & 25 October, 2013, Arni University, Himachal Pradesh.
14. Presented a paper entitled “Effect of Annealing on Physical Characteristics and Photoelectrochemical Response of Nanostructured ZnO”, International Conference on Environmental Science and Technology 2012 (ICEST 2012), Houston, USA

15. Participated in Workshop on “Virtual Power Lab” 18th February 2012, organized by K.P.E.C Technical Campus, Agra.

16. Presented a paper entitled “A Clean and Green Hydrogen Energy Production using Nanostructured ZnO and Fe-ZnO via Photoelectrochemical Splitting of water”, International Conference on Chemistry of Phytopotentials: Health, Energy and Environmental Perspectives (CPHEE 2011), Dayalbagh Educational Institute, Agra, India

17. Presented a paper entitled “Nanostructured ZnO and Fe-ZnO thin films for photoelectrochemical splitting of water”, International Conference on Nanoscience and Nanotechnology (ICNN 2011), Coimbatore, Tamil Nadu, India.

18. Participated in National Workshop o “Recent Trends in Environmental Chemistry” organized by Dayalbagh Educational Institute, Agra on 10th September 2011.

19. Presented a paper entitled “Synthesis and Characterization of Nanostructured ZnO for Solar Hydrogen Generation: Effect of Electrolyte pH on Photoelectrochemical Application”, in International Conference on Nanoscience and Technology in Chemistry, Health, Environment and Energy 2010 (NATCHEE 2010), Dayalbagh Educational Institute, Jan 2010, Dayalbagh, Agra, India.

20. Participated in Environment Meet – 2010 on the theme “My Dream of Agra as an Eco-City” organized by SPHEEHA, 5th December 2010.

21. Presented a paper entitled “Clean and renewable hydrogen energy production using nanostructured zinc oxide via photoelectrochemical splitting of water” in International Conference on Recent Advances in Environmental Protection (RAEP 2009), St. John College, Dec 17-19, Agra, India.

22. Presented a paper entitled “Ultrasonic assisted Synthesis of nanostructured SrTiO₃ for Photoelectrochemical Splitting of water”, in International Conference on Multifunctional oxide Materials, Himachal Pradesh University (ICMOM 2009), April 16th-18th, 2009, Shimla, India.

23. Participated in seminar on “Facing the Challenges of Climate Change and Global Warming in Agra (CG-GWA-09) organized by SPHEEHA on March 1, 2009, Agra.

24. Participated in National Workshop o “Recent Trends in Chemistry” organized by Dayalbagh Educational Institute, Agra on 2nd February 2008.

25. Participated in International Workshop on Biohydrogen Production Technology (IWBT 2008), 7 to 9, 2008, IIT Kharagpur, India.

Training Programs

1. Participated in ten days short term program on “New Education Policy – 2020” held from 2-11 February 2023 in J S University, Shikohabad.

2. Participated in training entitled as “Assessment Based Examination System” held on 16 December 2021 at Lovely Professional University, Punjab.

3. Participated in two days training program on “Energy Conservation Building Code 2017”, 5-6 December 2020 organized by Punjab Energy Development Agency.

4. Participated in training entitled as “Innovative Teaching Methods” held on 11 July 2020 at Lovely Professional University, Punjab.

5. Participated in three day “Hands on Training in Powder XRD, Diffrac. Eva and Diffrac. TOPAZ” organized by Bruker India Pvt Ltd in Lovely Professional University, Punjab, 23-25 September 2019.
6. Conducted training on “University Management System” on 1 January 2018 at Lovely Professional University, Punjab.
7. Participated in training entitled as “How to write successful Research Grant Proposal” held on 20 January 2018 at Lovely Professional University, Punjab.
8. Participated in training entitled as “Schemes of Funding For Faculty Training (SOFFFT)” held on 21 April 2018 at Lovely Professional University, Punjab.
9. Participated in training entitled as “Mentoring Students” held on 19 August 2017 at Lovely Professional University, Punjab.
6. Participated in training entitled as “Training on Academic Manual for Teachers, Instruction Plan and Lab Manual” held on 16 June 2017 at Lovely Professional University, Punjab.
10. Participated in training entitled as “Instrumental and Analytical Techniques” held on 1 April 2017 at Lovely Professional University, Punjab.
11. Participated in training entitled as “Workshop on Patent search on Xlpat” held on 14 January 2017 at Lovely Professional University, Punjab.
12. Participated in training entitled as “Collaborative Research” held on 3 September 2016 at Lovely Professional University, Punjab.
13. Participated in training entitled as “Curriculum Development Review - Instruction Plan Creation III” held on 16 December 2016 at Lovely Professional University, Punjab.
14. Participated in training entitled as “LiveLegendary- A Perspective For Better Teaching And A Better Life” held on 6 February 2016 at Lovely Professional University, Punjab.
15. Participated in training entitled as “Creative Teaching And Learning Strategies” held on 6 February 2016 at Lovely Professional University, Punjab.
16. Participated in training entitled as “Course Coordination: an Effective Tool for Improvement in Curriculum” held on 7 November 2015 at Lovely Professional University, Punjab.
17. Participated in training entitled as “Workshop on Design of MCQs” held on 5 September 2015 at Lovely Professional University, Punjab.
18. Participated in training entitled as “Time Management” held on 7th November 2015 at Lovely Professional University, Punjab.

Coordinator of Conference/Workshop

1. Organized 106th Indian Science Congress in 2019 in a capacity of “Local Sectional Secretary – Environmental Sciences” held from 3rd to 7th January 2019.
2. Conducted National Workshop on X-Ray Diffraction: Theory and Practice, 2019 in Lovely Professional University.

3. Organized Summer Science Camp from 4th – 27th June 2018 in a capacity of Faculty Coordinator in Lovely Professional University.
4. Organized “National Workshop on Environmental Sciences” held on 10th January 2018 as a Program Coordinator at Lovely Professional University, Punjab.
5. Organized “National Workshop on Scientoonics” held on 25th November 2017 as a Program Coordinator at Lovely Professional University, Punjab.
6. Organized a National Conference on “Recent Trends in Chemical & Environmental Sciences”, 27-28 February 2014 as a Convener at Arni University.

References

1. Prof. Ramesh Thakur

Dean

Department of Chemistry

Himachal Pradesh University, Shimla, HP

Email: drthakurchem@gmail.com

Mob: 9876071296

2. Prof. Sahab Dass

Head

Department of Chemistry

Dayalbagh Educational Institute, Agra, UP

Email: drsahabdas@gmail.com

3. Van-Huy Nguyen

Faculty of Applied Sciences,

Ton Duc Thang University,

Ho Chi Minh City, Vietnam.

Nguyenvanhuy20@duytan.edu.vn

Permanent Address:

Dr. Pushpendra Kumar

s/o. Shri. Yogendra Pal Singh

887, Shambhunagar, Shikohabad

Dist: Firozabad, Uttar Pradesh

Pincode: 205135

Mob: 8894900779

Email: Pushpendra.kmr1@gmail.com